AQ1.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.

5.1 Level 1:

1 point

Choose the set of correct options

☐ If AA is a real $3 \times 33 \times 3$ matrix, then $det(A) = det(A^T)$det(A) = det(AT).

☐ If $A = [a_{ij}]$A = [aij] is a real $4 \times 44 \times 4$ matrix, then the order of the sub matrix obtained by deleting the ii-th row and jj-th column of AA is $4 \times 44 \times 4$.

☐ If $A = [a_{ij}]$A = [aij] is a real $4 \times 44 \times 4$ matrix, then the order of the sub matrix obtained by deleting the ii-th row and jj-th column of AA is $3 \times 33 \times 3$.

☐ If AA is a real $3 \times 33 \times 3$ matrix, then the orders of all possible square sub matrices of AA are $1 \times 1, 2 \times 2,$ and $3 \times 31 \times 1, 2 \times 2,$ and 3×3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If AA is a real $3 \times 33 \times 3$ matrix, then $det(A) = det(A^T)$det(A) = det(AT).

If $A = [a_{ij}]$A = [aij] is a real $4 \times 44 \times 4$ matrix, then the order of the sub matrix obtained by deleting the ii-th row and jj-th column of AA is $3 \times 33 \times 3$.

If AA is a real $3 \times 33 \times 3$ matrix, then the orders of all possible square sub matrices of AA are $1 \times 1, 2 \times 2,$ and $3 \times 31 \times 1, 2 \times 2,$ and 3×3.

1 point

Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix}$A = $\begin{bmatrix} 3 & 2 & 2 & 2 \\ 3 & 1 & 2 & 2 \\ 1 & 2 & 2 & 1 \end{bmatrix}$ be a $3 \times 33 \times 3$ matrix. Which of the following is(are) correct?

[Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

☐ $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) − 2 × det ([2221]) + 2 × det ([2231])

☐ $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) + 2 × det ([2122]) + 2 × det ([2231])

☐ $det(A) = -2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) + 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$det(A) = −2 × det ([2121]) + 3 × det ([3221]) − 2 × det ([3221])

☐ $det(A) = 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) - 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$det(A) = 2 × det ([2121]) − 3 × det ([

]) + 2 × det ([3221])

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = 3 \times det([\begin{matrix}32\\11\end{matrix}]) - 2 \times det([\begin{matrix}22\\21\end{matrix}]) + 2 \times det([\begin{matrix}23\\21\end{matrix}])$det(A) = 3 × det ([3121]) − 2 × det ([2221

]) + 2 × det ([2231])

$det(A) = 3 \times det([\begin{matrix}32\\11\end{matrix}]) + 2 \times det([\begin{matrix}22\\12\end{matrix}]) + 2 \times det([\begin{matrix}23\\21\end{matrix}])$det(A) = 3 × det ([3121]) + 2 × det ([2122

]) + 2 × det ([2231])

$det(A) = -2 \times det([\begin{matrix}22\\11\end{matrix}]) + 3 \times det([\begin{matrix}32\\21\end{matrix}]) - 2 \times det([\begin{matrix}32\\21\end{matrix}])$det(A) = −2 × det ([2121]) + 3 × det ([3221

]) − 2 × det ([3221])

1 point
If all the elements of a $3 \times 3$3 × 3 real matrix AA are the same, then which of the following is (are) correct?

☐ Determinant of matrix AA is 00.

☐ Determinant of matrix AA cannot be determined from the given information.

☐ Determinant of matrix AA will be the sum of the elements of a row.

☐ Determinant of matrix $A + A^T$A + AT is 00, where A^TAT denotes the transpose of AA.

☐ Determinant of matrix $A + A^T$A + AT cannot be determined from the given information, where A^TAT denotes the transpose of AA.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Determinant of matrix AA is 00.
Determinant of matrix $A + A^T$A + AT is 00, where A^TAT denotes the transpose of AA.

1 point
Choose the set of correct options

☐ If two rows are the same in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.

☐ If two columns are the same in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.

☐ If one row is a non-zero multiple of another row in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is not zero.

☐ If one column is a non-zero multiple of another column in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If two rows are the same in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.
If two columns are the same in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.
If one column is a non-zero multiple of another column in a $3 \times 3$3 × 3 real matrix, then the determinant

of that matrix is zero.

Let $A = \begin{bmatrix} 1010 \\ 0101 \\ 1001 \\ 0001 \end{bmatrix}$A = $\begin{bmatrix} & & & \\ & & & \\ & & & \\ 1010010010000111 & & & \end{bmatrix}$ be a matrix of order $4 \times 4$4 × 4. Find $det(A)$det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -1
1 point

5.2 Level 2:

1 point
Let $A, B,$ A, B, and CC be $3 \times 3$3 × 3 real matrices. Which of the following options is (are) correct?

☐ $det(ABC) = det(A)det(B)det(C)$det(ABC) = det(A)det(B)det(C)

☐ $det(A^3) = det(A)^3$ det(A3) = det(A)3

☐ $det(A + B + C) = det(A) + det(B) + det(C)$det(A + B + C) = det(A) + det(B) + det(C)

☐ $det(AB^T) = det(A)\ det(B)$det(ABT) = det(A) det(B), where B^T BT denotes the transpose of B B.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(ABC) = det(A)det(B)det(C)$det(ABC) = det(A)det(B)det(C)
$det(A^3) = det(A)^3$ det(A3) = det(A)3
$det(AB^T) = det(A)\ det(B)$det(ABT) = det(A) det(B), where B^T BT denotes the transpose of B B.
1 point
Let $A = [\delta_{ij}]$A = [δij] be a matrix of order $3 \times 3$3 × 3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i > j \\ j & \text{if } i \leq j \end{cases}$δij = {0jif i > jif i ≤ j

Choose the set of correct options

[Hint: Write down matrix AA first.]

☐ AA is an upper triangular matrix.

☐ AA is a lower triangular matrix.

☐ $det(A) = 6$det(A) = 6

☐ $det(A) = 3$det(A) = 3

No, the answer is incorrect.
Score: 0

Accepted Answers:
AA is an upper triangular matrix.
$det(A)$ = 6det(A) = 6

1 point

Let $A = [\delta_{ij}]$A = [δij] be a matrix of order $3 \times 3$3 × 3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i < j \\ 2 & \text{if } i \geq j \end{cases}$δij = {02if i < jif i ≥ j

Choose the set of correct options

- [ ] AA is an upper triangular matrix.
- [ ] AA is a lower triangular matrix.
- [ ] $det(A)$ = 2det(A) = 2
- [ ] $det(A)$ = 8det(A) = 8

No, the answer is incorrect.
Score: 0
Accepted Answers:
AA is a lower triangular matrix.
$det(A)$ = 8det(A) = 8

Suppose $A = \begin{bmatrix} 4 & -1 & 3 \\ 2 & 0 & 1 \\ 3 & -2 & 0 \end{bmatrix}$A = $\begin{bmatrix} 4 & 2 & 3 & -1 \\ 0 & -2 & 3 & 1 \\ 0 & 0 & -2 & 3 \end{bmatrix}$ and C_{ij} Cij is the (i, j)(i, j)-th cofactor of the matrix AA.

Let BB be $3 \times \times 3$ matrix where (i, j)(i, j)-th entry is equal to C_{ij} Cij. Find $det(B)$det(B).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 49

1 point

Suppose AA and BB are two $3 \times \times 3$ matrices such that $det(A)$det(A) = 4 and $B = 3A$B = 3A. Find the value of $\sqrt[3]{det(A^2 B)}$3 det(A2B)

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No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 12

1 point

Check Answers

Your score is: 0/10